OPEN SOURCE BLOCKCHAIN

DECENTRALIZATION AND INNOVATION

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OUTLINE

- Introduction 01
- Blockchain 02
- Bitcoin 03
- Web3 04
- Muon Network 05

INTRODUCTION

Open source software is publicly accessible, allowing anyone to view, modify, and distribute the code.

- **Transparency:** Anyone can view and understand the code.
- **Collaboration:** Developers worldwide can contribute, bringing diverse perspectives.
- **Innovation:** Continuous improvements and new features are added rapidly.
- **Community:** Developers share knowledge and support each other.

OPEN SOURCE EXAMPLES

Programming Languages

- Python, Go

Operating Systems

- Linux

Blockchain Projects

- Bitcoin: The first decentralized cryptocurrency.
- Ethereum: A platform for decentralized applications.
- Muon Network: Integrates with blockchain and Web3, promoting innovation and community.

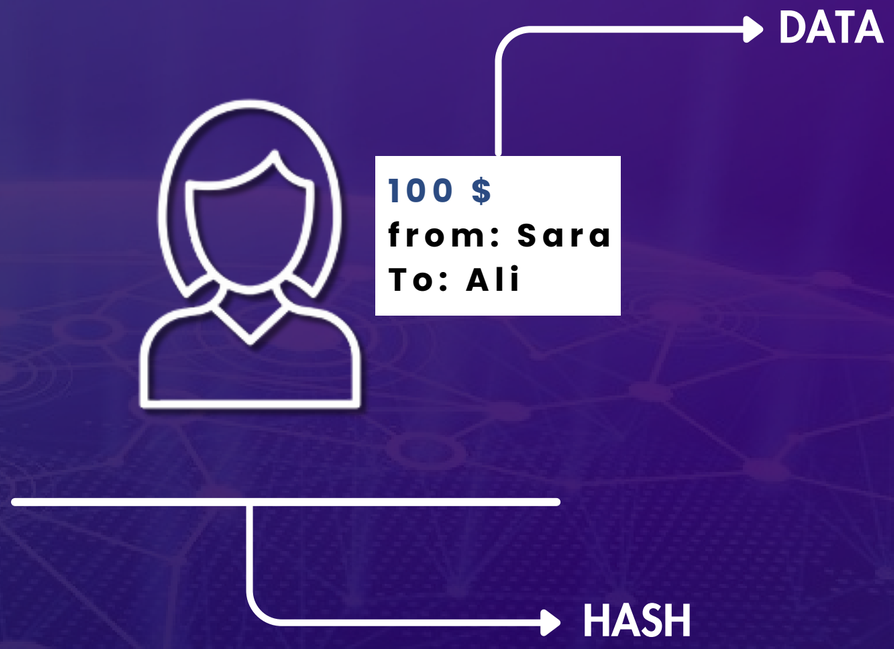
BLOCKCHAIN

Blockchain is a decentralized digital ledger that records transactions across many computers in such a way that the registered transactions cannot be altered retroactively.

- **1991:** Stuart Haber and W. Scott Stornetta conceptualized a cryptographically secure chain of blocks.
- **2008:** Satoshi Nakamoto introduced Bitcoin and the blockchain as its underlying technology in the whitepaper "Bitcoin: A Peer-to-Peer Electronic Cash System."
- **2009:** Bitcoin was launched, marking the first practical implementation of blockchain technology.

EXAMPLE





EXAMPLE





100 $
from: Diana
To: Sara

Previous HASH

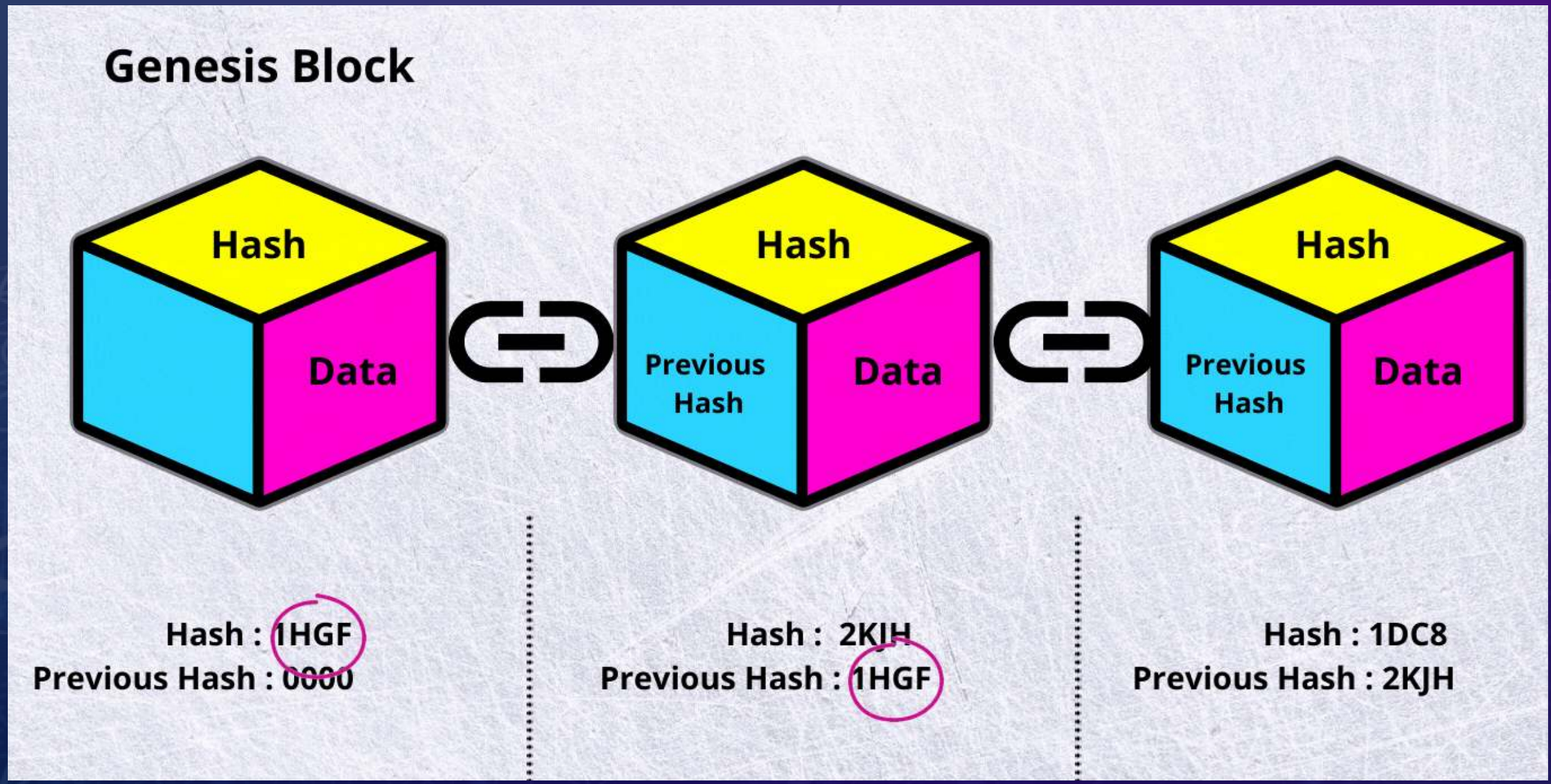
DATA



100 $
from: Sara
To: Ali

HASH

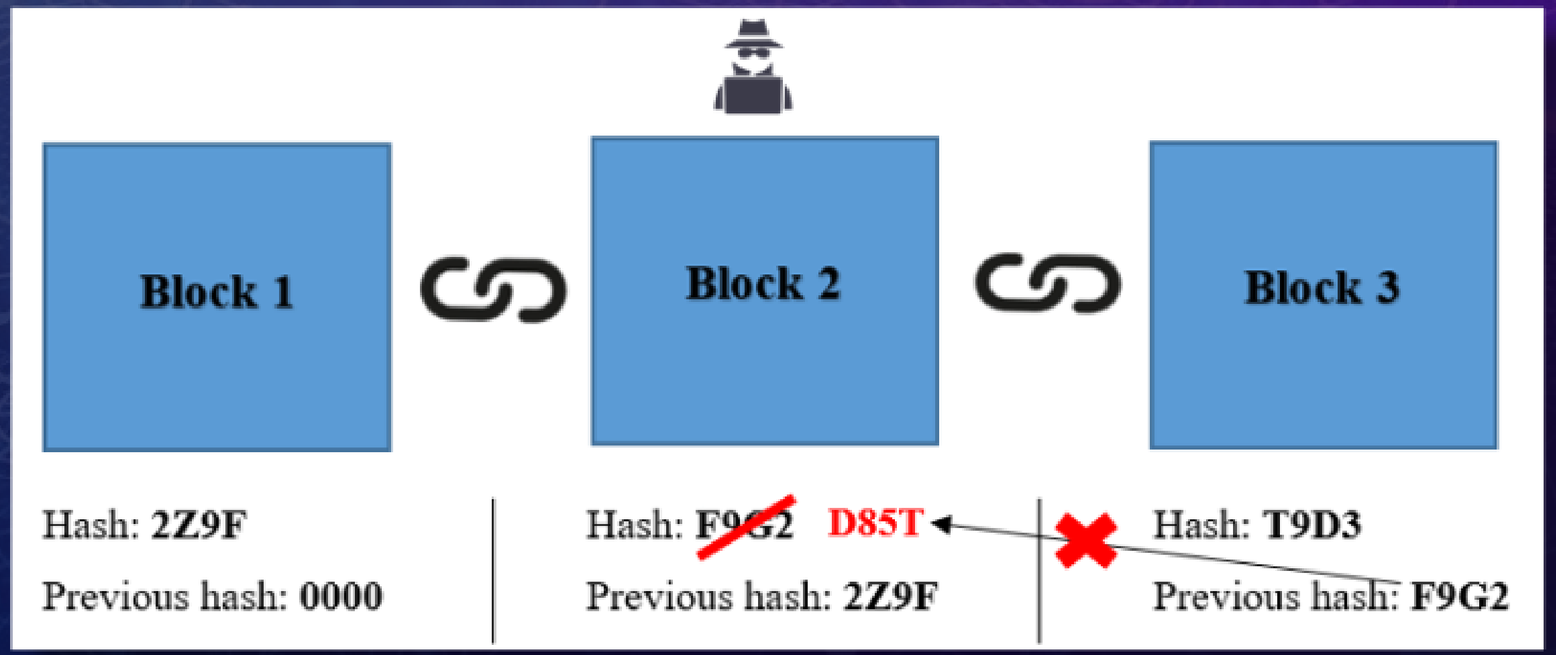
BLOCKCHAIN


Genesis Block
Hash
Data
Hash
Previous Hash
Data
Hash
Previous Hash
Data
Hash : 1HGF
Previous Hash : 0000
Hash : 2KJH
Previous Hash : 1HGF
Hash : 1DC8
Previous Hash : 2KJH

HASH

Unique hash for each block, changes with data alteration.

- **Hash Security:** Links to previous block's hash, ensuring integrity.
- **Role of Nonce:** Random number making hash generation complex.
- **Mining Process:** Miners compete to find nonce, add block, earn reward.



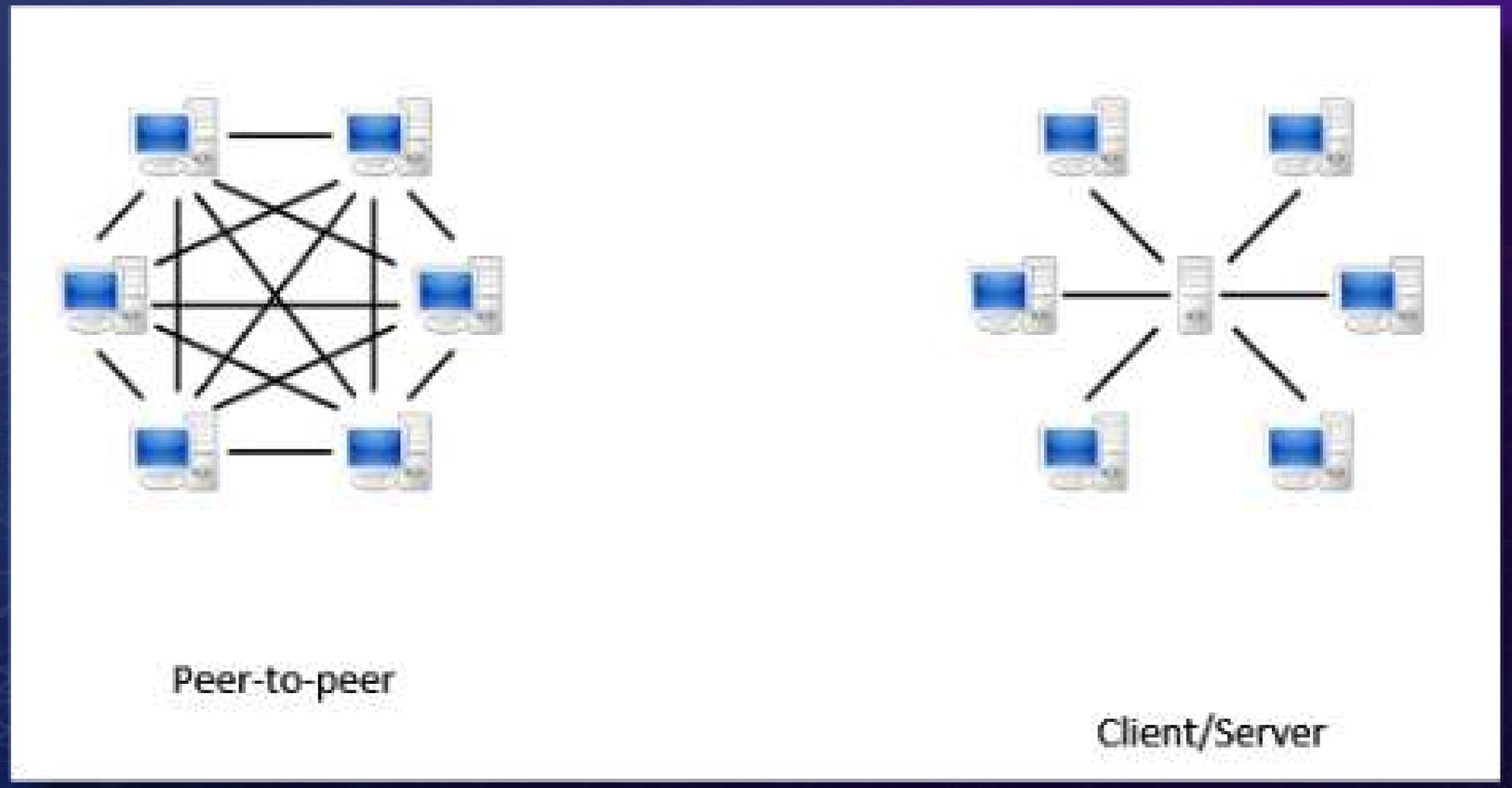
BLOCKCHAIN

1	2	3	4	5
CREATE TRANSACTION	**BROADCAST TRANSACTION**	**NODES VERIFY TRANSACTION**	**TRANSACTION APPROVAL**	**MINING PROCESS**
Sarah creates a transaction to send Bitcoin to Ali with her signature.	The transaction is broadcast to all nodes in the blockchain network.	Nodes verify the transaction's signature and Sarah's balance.	Valid transactions are approved and added to a new block.	Miners begin mining, changing the nonce to find a valid hash.

BLOCKCHAIN

6	7	8	9	10
FIND VALID HASH	**NODES VALIDATE NEW BLOCK**	**ADD BLOCK TO BLOCKCHAIN**	**ALI VIEWS TRANSACTION**	**COMPLETE TRANSACTION**
The first miner to find a valid hash sends the new block to the network.	Nodes validate the new block's hash and nonce.	The new block is added to the blockchain, updating all nodes.	Ali sees the transaction and his updated balance.	The transaction is completed and receives more confirmations over time.

P2P NETWORKS



OPEN SOURCE BLOCKCHAIN PROJECTS

Most blockchain tools are open source, promoting transparency and collaboration. Here are some notable

- **Bitcoin (Bitcoin GitHub Repository) :**
  - The first decentralized cryptocurrency.
- **Muon Network (Muon Network GitHub Repository):**
  - Decentralized network integrating with blockchain and Web3 applications.
- **Hyperledger Fabric (Hyperledger Fabric GitHub Repository):**
  - Framework for developing blockchain-based supply chain solutions.
- **Medicalchain (Medicalchain GitHub Repository):**
  - Uses blockchain to securely store and share electronic health records.
- **Ethereum (Ethereum GitHub Repository):**
  - Platform for decentralized applications and smart contracts.

BITCOIN

Bitcoin is a decentralized digital currency and open-source project. Transactions are secured by cryptography and recorded on a public ledger called the blockchain.

Published by: Satoshi Nakamoto, 2008

Purpose: Enable peer-to-peer transactions without relying on a central authority

Main Concept: Decentralized digital currency, secured by cryptography

Key Features

Decentralization: No central authority or intermediaries

Security: Transactions secured by cryptographic algorithms

Transparency: All transactions are publicly recorded on a blockchain

WEB3


Web Evolution from 1.0 to 3.0
Web1.0
Web2.0
Web3.0
WIKIPEDIA
The Free Encyclopedia
LINE
You
Tube

WEB3

Data Ownership

- Users own and control their data.
- No central authority.

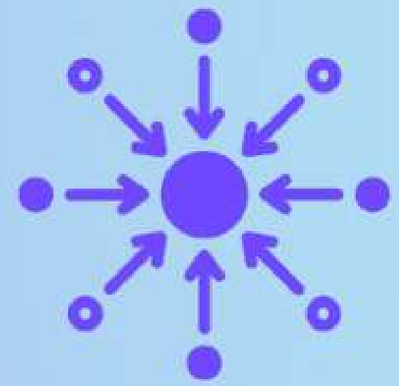
Challenges

- Potential for violent content.
- Need for effective moderation.

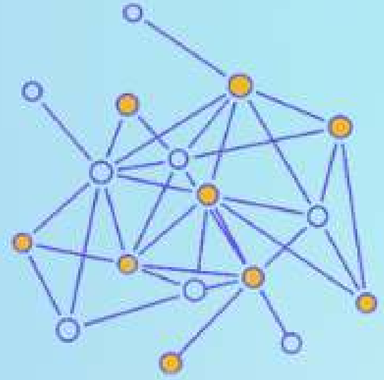
Solutions

- DAO & Smart Contracts
  - DAO: Community votes on content decisions.
  - Smart Contracts: Automated enforcement of rules.

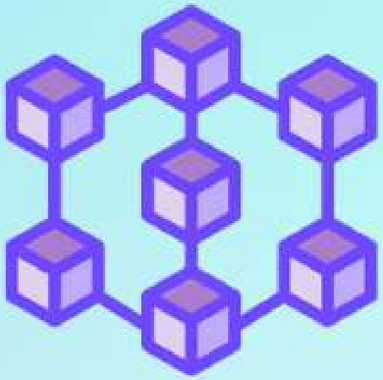
WEB3



WEB 1.0
Read-Only: 1990-2004



WEB 2.0
Read-Write: 2004-now



WEB 3.0
Read-Write-Own: 2014-?

MUON NETWORK

The Muon Network is a decentralized integration layer designed to connect blockchain and Web3 applications. It facilitates secure and scalable interoperability between different blockchains and traditional web systems, enhancing the functionality and reach of decentralized applications

- **Use Cases:**
  - **Oracles:** Provides real-world data to smart contracts.
  - **Cross-Chain Bridges:** Enables seamless asset and data transfers across different blockchains.
  - **Data Feeds:** Supplies reliable and real-time data to decentralized applications (dApps).
- **Future Prospects:**
  - **Expansion:** Aims to integrate with more blockchains and traditional systems.
  - **Innovation:** Continuously developing new features to enhance functionality and user experience.

ROAD MAP

- **Introduction:** Learn blockchain basics.
- **Use Cases:** Explore applications.
- **Wallet and Testnet:** Set up a wallet and practice with testnet coins.
- **Smart Contracts:** Write and deploy smart contracts using Solidity.
- **Run a Node:** Set up and interact with nodes using C++ and Go.
- **Develop dApps:** Build and deploy decentralized applications using JavaScript and Web3.js.
- **Advanced Topics:** Study consensus, scalability, and security using Rust and Python.
- **Get Involved:** Join communities, contribute to projects, and stay updated.

REFERENCES

- https://github.com/bitcoin
- https://developer.bitcoin.org/devguide/
- https://developer.bitcoin.org/reference/block_chain.html
- https://pion.muon.net/
- https://github.com/muon-protocol

THANK YOU

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